

Muhammad Ali Waris Khan

AI & Backend Engineer — LLM & RAG Developer — Python Backend — Data Analytics

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 github.com/maw-khan —  LinkedIn —  Portfolio

Professional Summary

AI & Backend Engineer specializing in Retrieval-Augmented Generation (RAG), Large Language Models, Python backend development, REST APIs, PostgreSQL database systems, and data analytics solutions.

Experienced in building and deploying AI applications using Python, FastAPI, LangChain, Gemini AI, FAISS, PostgreSQL, Power BI, SQL, and modern data workflows.

Skilled in designing RAG pipelines, backend services, relational database models, interactive dashboards, and business intelligence solutions.

Strong foundation in async Python, API development, document intelligence, and production-oriented software engineering.

Technical Skills

Programming: Python, SQL, Async Python (Async/Await), Type Hints

Backend Engineering: FastAPI, REST APIs, HTTP, JSON, PostgreSQL, API Development, Exception Handling

LLM Engineering: RAG Pipelines, LangChain, LCEL, Gemini AI, Prompt Engineering, Vector Search, Semantic Retrieval

Databases & Analytics: PostgreSQL, Database Design, Schema Modeling, Star Schema, Data Cleaning, Power Query, DAX, Power BI Dashboards

Retrieval & AI: FAISS, Embeddings, Document Chunking, MMR Retrieval, Cross Encoder Reranking

Software Engineering: Git, GitHub, pytest (basic exposure), Debugging, Modular Architecture, Reusable Components

Deployment & Tools: Streamlit, FastAPI Deployment, Postman, VS Code

Nice to Have Exposure: LangGraph, OCR & Document Extraction, Cloud Concepts (AWS/Azure)

Professional Experience

Lab Engineer (Academic) — NUST, Islamabad

January 2025 – February 2026

- Assisted faculty members during laboratory sessions and practical coursework.
- Guided students in programming, debugging, and technical problem solving.
- Supported software setup, technical demonstrations, and computing resources.

Selected Projects

1. AI Resume Analyzer with ATS Scoring

(Python, FastAPI, Gemini AI, NLP, PyMuPDF)

- Built an AI-powered resume analysis platform using LLM-based evaluation and keyword matching.
- Developed PDF extraction and preprocessing pipelines with validation and exception handling.
- Created modular backend components and deployed application using FastAPI-based architecture.

2. YouTube Video RAG Chatbot

(Python, FastAPI, LangChain, Gemini AI, FAISS)

- Developed a Retrieval-Augmented Generation system for querying YouTube transcripts.
- Implemented transcript extraction, chunking, embeddings, vector search, and contextual answering.
- Built backend APIs for document processing and conversational interaction.

3. Urban Basket Sales Analytics Platform

(PostgreSQL, SQL, Power BI, DAX)

- Designed and loaded sales datasets into PostgreSQL database systems.
- Created relational data models and optimized database relationships for analytics workflows.
- Developed interactive Power BI dashboards using Power Query and DAX measures to analyze sales performance, revenue trends, and business KPIs.

4. MetroCare Healthcare Analytics Dashboard

(PostgreSQL, SQL, Power BI, Data Modeling)

- Integrated and transformed multiple CSV datasets into a structured PostgreSQL database.
- Designed analytical data models for hospital operations and patient insights.
- Built interactive dashboards covering appointments, cancellations, insurance analysis, patient categories, and revenue-related metrics.

Education

Masters in Artificial Intelligence

College of Electrical and Mechanical Engineering, NUST, Islamabad